

Computing Skills and Python

Student Number: 760072671

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Assignment Title: Github Coding Task and Short Report

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Short Report

Introduction: The purpose of this assignment was to design and implement a hospital database with a clear relational structure, and to query the data using Structured Query Language (SQL). Entity Relationship Diagrams (ERDs) and pseudocode were used to plan the database and demonstrate learning and development in key skills required for this module. The final schema consists of four tables: Hospital, Doctors, Patients, and Prescriptions. This report summarises design choices, implementation process, SQL development, GitHub use, and AI-assisted learning undertaken during the assignment.

Database Design: During the early stages of the design process, I chose to use both Entity Relationship Diagrams (ERDs) and pseudocode to ensure that my planning was robust and that my understanding of relational database concepts was clear. I opted to use tools familiar to me, such as Microsoft Excel, to keep the planning focused on core principles and practical application.

The ERD development involved creating a visual representation of the planned tables and their relationships. Using Excel allowed me to map out one-to-many relationships clearly, such as Hospital to Doctors and Doctor to Patients, and to confirm that each table had a clearly defined purpose. This visual planning helped me identify unnecessary fields early on. Although the CSV files contained additional data, I realised during implementation that this information was not required for the assignment, and removing it resulted in a cleaner, more efficient schema.

Pseudocode played an important role in planning the MySQL implementation. Writing out the logical steps before coding allowed me to test my understanding of table creation, data loading, and query structure without risking errors or corrupting the database. This process also helped me refine my approach once I began working directly in MySQL, and I updated the pseudocode to reflect the final structure and relationships.

Finally, establishing referential integrity from the outset was essential. By defining clear primary keys and foreign keys, I ensured that the relationships between tables were consistent and reliable. This gave me confidence that the SQL queries executed later in the project would return accurate and meaningful results.

Implementation in MySQL: The implementation stage took longer than expected, largely due to issues that emerged once I began loading data into MySQL. I initially created the tables based on my first draft of the pseudocode, and at that point I was confident that the columns, data types, and relational structures were correctly defined.

Difficulties arose when importing the CSV files. The files were not immediately accessible to MySQL, so I created copies within my root directory to enable loading. Once the data began importing, further issues became apparent. Several fields required changes to their data types; for example, the hospital accreditation field was initially designed as a single-character value ('Y' or 'N'), but the CSV files contained full text entries such as "Accredited" and "Not Accredited". To resolve this, I updated the field to VARCHAR(20) to ensure the data could be imported correctly.

I also identified data within the CSV files that was not required for the assignment. Removing unnecessary fields resulted in a cleaner and more focused schema, and reduced confusion during testing. After successfully loading the data, I tested the relationships between tables to confirm that foreign keys and joins behaved as expected.

A minor but persistent issue involved the naming of the hospital table. My pseudocode and ERD referred to "Hospitals", while my MySQL implementation used "Hospital". This inconsistency caused errors during query execution. Once I standardised the table name across all documents and updated the pseudocode and ERD accordingly, the implementation became stable and ready for SQL query development.

SQL Query Development: To develop the SQL queries, I began by identifying the specific objective of each task and mapping these to the appropriate SQL operations. There were six required queries, each targeting a different aspect of the database, and together they provided a comprehensive test of the relational structure.

The first three queries involved retrieving data using INNER JOIN to link related tables. Query one required listing doctors at a particular hospital, which I achieved by joining the Doctors and Hospital tables on HospitalID. Query two involved listing all prescriptions for a specific patient, again using an inner join between Patients and Prescriptions, with results ordered chronologically using ORDER BY. Query three followed a similar pattern, retrieving all prescriptions issued by a particular doctor.

Query four differed in that it required inserting new data into the database. I used the INSERT INTO statement to add a new patient and then ensured the patient was correctly linked to a doctor via the DoctorID foreign key. This query tested both data entry and referential integrity.

Query five required identifying the doctor who had issued the most prescriptions. I used COUNT(*) in combination with GROUP BY to calculate prescription totals for each doctor, and then ordered the results to identify the highest value. Query six involved determining which hospital had the most beds and then listing the doctors who worked there. This required ordering the Hospital table by bed count and linking the result back to the Doctors table.

Developing these queries involved considerable trial and error, particularly interpreting MySQL error messages and adjusting syntax or table references accordingly. However, by approaching each query systematically and testing outputs frequently, I was able to produce accurate results and deepen my understanding of SQL operations and relational database behaviour.

Use of GitHub: Once I had completed the SQL queries and documented my work, I created a dedicated GitHub repository to store and organise all components of the assignment. I structured the repository using clear folders to separate planning documents from SQL outputs, ensuring that each file was easy to locate and logically grouped.

I uploaded each document individually and included commit messages to describe the purpose of each file. The README file was updated throughout the process to reflect the evolving structure of the repository and to provide clear guidance for anyone reviewing the work. This included explanations of the database schema, the planning documents, and the SQL queries.

Although GitHub initially felt unfamiliar and somewhat clunky, repeated use made the workflow more intuitive. By the end of the assignment, I felt confident in creating repositories, organising files, and maintaining clear documentation through the README.

Use of AI: Throughout the planning, development, execution, and reflection stages of this assignment, I used Microsoft Copilot to support my learning and troubleshoot issues. AI was used to clarify concepts, identify gaps in my understanding, and provide feedback on my planning documents. At all times, I ensured that I understood the material fully and made independent decisions about the database design, SQL implementation, and documentation.

I followed the University of Exeter's guidance on the ethical use of generative AI, ensuring that all work submitted is my own. No code or text was copied directly from AI outputs; instead, I manually wrote all SQL queries, pseudocode, and documentation after reviewing and interpreting the guidance provided. I also consulted the module lead to confirm that my use of AI was appropriate and transparent.

Copilot does not provide exportable or shareable links to chat transcripts, so it was not possible to include direct hyperlinks to AI conversations. However, all AI-assisted reasoning has been incorporated into the submitted documents, and the chats remain stored locally should they be required for review.

Reflection & Conclusion: Throughout this assignment I have developed a much stronger understanding of pseudocode, SQL, Entity Relationship Diagrams, and GitHub. At the outset, many of these concepts were unfamiliar to me, but through sustained practice and problem-solving I am now able to use these tools with increasing confidence. The process has

highlighted the importance of careful planning, iterative development, and systematic testing when working with relational databases.

There were moments of frustration, particularly when encountering repeated error messages or technical issues, and my perfectionist tendencies made it challenging at times to work through problems patiently. However, these challenges were valuable learning experiences. They helped me develop resilience, improve my troubleshooting skills, and gain a clearer understanding of how MySQL behaves when data types, relationships, or table structures are not fully aligned.

The skills gained through this assignment will be directly applicable to future work within the module and across the MSc programme. I now feel more confident in approaching database design tasks, writing SQL queries, and using GitHub to organise and document technical work. The completed project demonstrates that I have met the Intended Learning Outcomes (ILOs) for this assessment and that I am progressing appropriately in my technical development at this stage of the programme.

In conclusion, the assignment has provided a solid foundation in relational database design and SQL and has strengthened my ability to plan, implement, and document technical work effectively. The experience has been challenging but ultimately rewarding, and it has equipped me with skills that I will continue to build on throughout the remainder of the MSc.